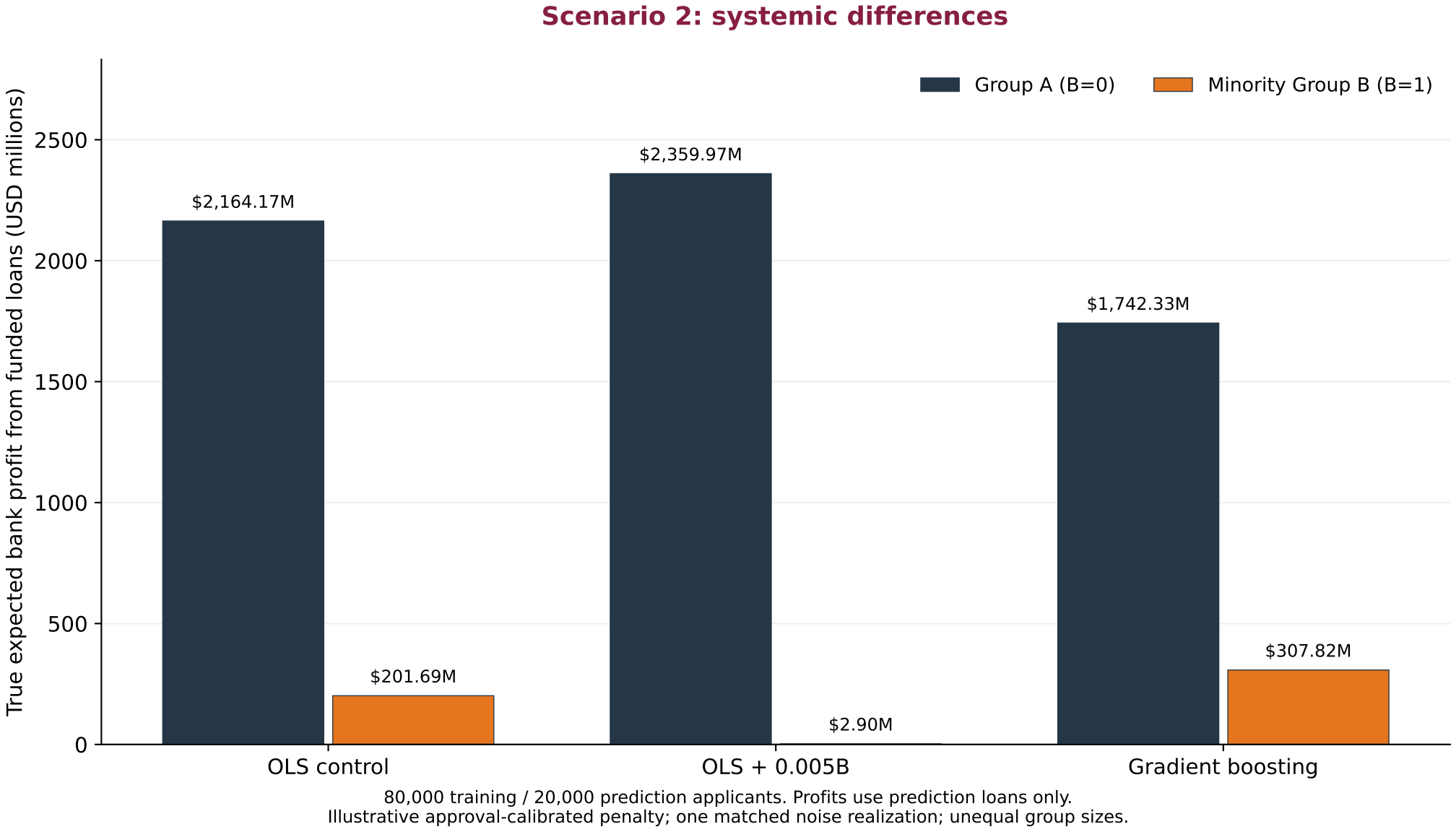
Scenario 2: systemic differences
Group A (B=0)
Minority Group B (B=1)
True expected bank profit from funded loans (USD millions)
0
500
1000
1500
2000
2500
$2,164.17M
$201.69M
$2,359.97M
$2.90M
$1,742.33M
$307.82M
OLS control
OLS + 0.005B
Gradient boosting
80,000 training / 20,000 prediction applicants. Profits use prediction loans only.
Illustrative approval-calibrated penalty; one matched noise realization; unequal group sizes.